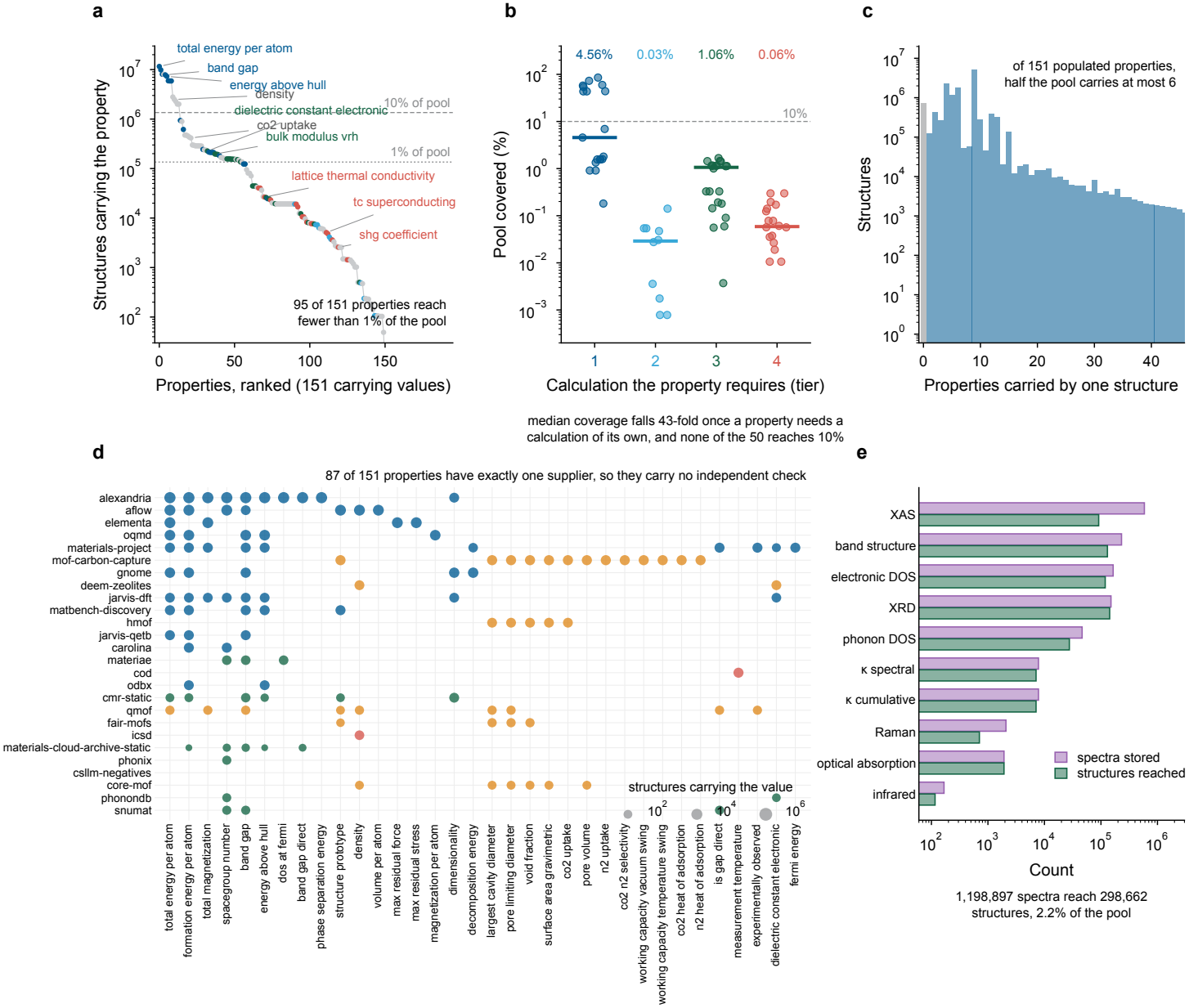

a
total energy per atom
band gap
energy above hull
density
dielectric constant electronic
co2 uptake
bulk modulus vrh
lattice thermal conductivity
tc superconducting
shg coefficient
10% of pool
1% of pool
95 of 151 properties reach fewer than 1% of the pool
Structures carrying the property
Properties, ranked (151 carrying values)
b
4.56%
0.03%
1.06%
0.06%
10%
Pool covered (%)
Calculation the property requires (tier)
median coverage falls 43-fold once a property needs a calculation of its own, and none of the 50 reaches 10%
c
of 151 populated properties, half the pool carries at most 6
Structures
Properties carried by one structure
d
87 of 151 properties have exactly one supplier, so they carry no independent check
alexandria
aflow
elementa
oqmd
materials-project
mof-carbon-capture
gnome
deem-zeolites
jarvis-dft
matbench-discovery
hmof
jarvis-qetb
carolina
materiae
cod
odbx
cmr-static
qmof
fair-mofs
icsd
materials-cloud-archive-static
phonix
csllm-negatives
core-mof
phonondb
snumat
total energy per atom
formation energy per atom
total magnetization
spacegroup number
band gap
energy above hull
dos at fermi
band gap direct
phase separation energy
structure prototype
density
volume per atom
max residual force
max residual stress
magnetization per atom
dimensionality
decomposition energy
largest cavity diameter
pore limiting diameter
void fraction
surface area gravimetric
co2 uptake
pore volume
n2 uptake
co2 n2 selectivity
working capacity vacuum swing
working capacity temperature swing
co2 heat of adsorption
n2 heat of adsorption
is gap direct
measurement temperature
experimentally observed
dielectric constant electronic
fermi energy
structures carrying the value
e
XAS
band structure
electronic DOS
XRD
phonon DOS
κ spectral
κ cumulative
Raman
optical absorption
infrared
spectra stored
structures reached
Count
1,198,897 spectra reach 298,662 structures, 2.2% of the pool